

ARMAAN SINGH

Surrey BC | asa560@sfu.ca | +1-(604)-841-5141 | [linkedin.com/in/armaan05](https://www.linkedin.com/in/armaan05) | Availability: 12 months

Education

Bachelor of Applied Science, Engineering (CGPA: 4.02/4.33)

Sep 2023 to Present

- Simon Fraser University – Burnaby, BC

Skills

- *Software & Firmware:* C/C++, VHDL, TCL (Scripting), MATLAB, Linux/Embedded OS, Assembly (RISC-V ISA), Linux.
- *Hardware & Verification:* ModelSim/Quartus, LTspice, ngSpice, Simulink, Oscilloscopes, Function Generators, DMM, VS Code.
- *Professional & Analytical:* Debugging & Analysis, Technical Documentation, Project Execution, Team Collaboration.

Experience

AI Contributor / Data Annotator - DataAnnotation – Remote

Jan 2026 to Present

- Evaluate outputs from advanced Large Language Models (LLMs) to improve response accuracy, safety, thoroughness, instruction-following, code quality, and helpfulness.
- Execute agentic coding tasks focusing on debugging, testing, and optimizing hardware/software interaction code to ensure seamless system integration and repair.

Teaching Assistant - Kumon – Surrey, BC

Sep 2021 to Sep 2023

- Conducted timely and accurate assessments of student work, providing constructive feedback aligned with instructor guidelines to promote academic growth.
- Documented student progress, enabling tailored support to enhance individual learning outcomes.
- Utilized communication and interpersonal skills to effectively explain complex mathematical and reading concepts to students of varying learning styles.

Projects

RISC-V Processor Design & Architecture (ENSC 350)

April 2026

- Architected a custom RISC-V core, implementing synthesized arithmetic, logic, and shifter hardware modules.
- Authored comprehensive technical documentation detailing internal microarchitectural decisions and functional specifications.
- Automated functional verification and timing analysis routines using ModelSim TCL scripting for custom ALU blocks.
- Evaluated Cyclone IV and Arria II FPGA targets to optimize power consumption, resource utilization, and overall execution speed.

Matrix Arcade - Real-Time Embedded Multi-Player Game Console (ENSC 351)

December 2025

- Constructed a complete embedded console using BeagleY-AI, involving UDP, HAL, and electronic hardware integration.
- Optimized hardware updates for a 64x32 LED matrix, focusing on signal stability and low-latency response troubleshooting for seamless play.
- Calibrated system components to maintain high-fidelity output and ensured all mechanical connections were secured for long-term durability.

Analog Signal Processing Filter Design (ENSC 320)

August 2025

- Designed and built a physical 4th-order filter to suppress noise, performing manual signal processing calibration and precision hardware repair.
- Validated circuit reliability using oscilloscopes and signal generators, ensuring hardware operation remained within the allowed tolerances.
- Performed hands-on troubleshooting and component replacement to improve signal-to-noise ratios and stabilize filtered frequency responses.

Awards

- SFU President's and Dean's Honor Roll
- SFU Open Scholarship